4.13.46

EE25BTECH11042 - Nipun Dasari

Question:
A straight line L is perpendicular to the line $5x - y + 1 = 0$. The area of the triangle formed by L and the coordinate axes is 5. Find the equation of the line L.
Solution:
Let the given line be L_1 and the required perpendicular line be L. The equation of the given line L_1 is $5x - y = -1$. Using the matrix form $\mathbf{n}^\top \mathbf{x} = c$, its parameters are:

$$\mathbf{n}_1 = \begin{pmatrix} 5 \\ -1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad c_1 = -1 \tag{0.1}$$

Let the required line L have the equation $\mathbf{n}_2^\top \mathbf{x} = c_2$. Since the lines are perpendicular, the dot product of their normal vectors must be zero, so $\mathbf{n}_1^\top \mathbf{n}_2 = 0$.

Let's assume the normal vector for line L is $\mathbf{n}_2 = \begin{pmatrix} 1 \\ m \end{pmatrix}$. We can solve for m:

$$\mathbf{n}_1^\top \mathbf{n}_2 = \begin{pmatrix} 5 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ m \end{pmatrix} = 5 - m = 0$$
$$\implies m = 5 \tag{0.2}$$

Therefore, the normal vector for line L is $\mathbf{n}_2 = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$, and its equation is $x + 5y = c_2$.

The line L forms a triangle with the coordinate axes. We find the intercepts. For the x-intercept, set $y = 0$, which gives $x = c_2$. The point is $(c_2, 0)$. For the y-intercept, set $x = 0$, which gives $5y = c_2$, so $y = c_2/5$. The point is $(0, c_2/5)$.

The area of this triangle is given as 5. Using the area formula:

$$\text{Area} = \frac{1}{2}|(\text{x-intercept}) \cdot (\text{y-intercept})| = 5$$
$$\frac{1}{2}|(c_2)(c_2/5)| = 5$$
$$|\frac{c_2^2}{10}| = 5$$
$$c_2^2 = 50 \implies c_2 = \pm\sqrt{50} = \pm 5\sqrt{2} \tag{0.3}$$

There are two possible lines that satisfy the given conditions. Substituting the values of c_2 into the matrix equation $\mathbf{n}_2^\top \mathbf{x} = c_2$, the possible equations for line L are:

$$\begin{pmatrix} 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 5\sqrt{2} \tag{0.4}$$

$$\begin{pmatrix} 1 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -5\sqrt{2} \tag{0.5}$$